SURNAME, FIRST NAME: EBU6230 Class Test 1, 2020/21

BUPT ID: **QM ID:** TOTAL MARK (examiner's use)

Q1. In HSV colour model, saturation represents -

A: the colour's purity **B:** the dominant wavelength **C:** how light or dark the colour is **D:** None

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Q2. *True or False:* Coarse quantization often causes to appear false contours in an image.

A: True **B:** False

Q3. The range of wavelengths visible to the human eye is approximately:

A: 0-100nm **B:** 200-600nm **C:** 300-500nm **D:** 400-700nm

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Q4. **I**f a pixel is encoded by 7 bits, and 0 is black, then what value is white?

A: 1 **B:** 255 **C:** 127 **D:** 7

Q5. What type of transformations is represented by the following equation, where (x, y) are input coordinates, and (x', y') are output coordinates?

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 1 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

A: 2D Rotation **B:** 2D Scaling **C:** 2D Translation **D:** 2D Mirror

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Q6. Suppose that an image contains N pixels in total. If both the horizontal *and* vertical resolution is doubled, then how many pixels does the new image contain?

A: Same (i.e., N) **B:** 2N **C:** N^2 **D:** 4N

Q7. Which colour model does more closely correspond to the way humans perceive/see colours?

A: RGB **B:** HSV **C:** Both A & B **D:** None

Q8. Both rods and cones are found in the human visual system, where rods are responsible for _____.

A: Colour **B:** Purity of colour **C:** Brightness **D:** Intensity

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Q9. Both rods and cones are found in the human visual system, where cones are responsible for _____.

A: Colour **B:** Purity of colour **C:** Brightness **D:** Intensity

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Q10. State **TRUE** or **FALSE** for the following statement:

"*Histograms provide the spatial information of an image*".

A: TRUE **B:** FALSE

Q11. Averaging several images of the same view is useful for:

A: Dithering **B:** Edge detection **C:** Visualization **D:** Noise reduction

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Q12. Give the name(s) of the application of image and video processing: What technique is used to separate an object from background using histogram?

A: Histogram thresholding **B:** Histogram equalization **C:** Histogram **D:** None

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WRITE YOUR ANSWER UNDER EACH QUESTION:

Q13. Given a colour represented in CMY colour space as C = 0.6, M = 0.5, and Y = 0.7, what is its representation in the CMYK colour model? **[4 marks]**

Q14. Suppose that an image is subsampled by a factor of two. What happens to the size of the image? Now suppose that the size of each pixel is increased by the same factor; what problem might result? **[2 marks]**

Q15: Consider the following greyscale pixel-values:

11	44	221
15	0	89
25	9	3

Write a full PGM header for this data (Binary format), including a comment line. **[4 marks]**